



AGENT OS

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INTRODUCTION

- Internet - world wide information repository
- Resource sharing on a world wide basis
- Agents are new approach to the development of distributed client-server application



AGENT OS

- Supports agent-based client-server applications
- Behaves as a traditional server for a single host
- Entire network-provides an environment for distributed applications



AGENTS

- Object representation of distributed systems
- Contains both computational logic and state Information
- It can be a single algorithm or a complete application
- Agents are active and mobile



AGENT PARADIGM

- Classical architecture involves two parts:
 - Lower Level layer- Implementing communication protocol
 - Higher Level layer- Implementing algorithm
- Drawbacks
 - The complexity of the lower layer
 - Separation between these layers may not be clear



AGENT PARADIGM(CONTD..)

- Provide solution to above problems
- Distributed application is broken down into components –Agents
- Agents are then used to carry out the algorithms
- Agents behaves more like a centralized application
- AgentOS is responsible
 - transporting agents
 - communication between agent hosts



AGENT PARADIGM(CONTD..)

- Separation b/w communication protocol layer & application layer
- Changes in applications are localized to the application layer
- Changes in communication protocol do not have an effect on application



APPLICATION ACTIVITIES SUITED TO MOBILE AGENTS

- Distributed Data Collection
- Monitoring
- Information Delivery and 'Push'
- Negotiation
- Transaction
- Parallel Processing



THE AGENT LIFECYCLE

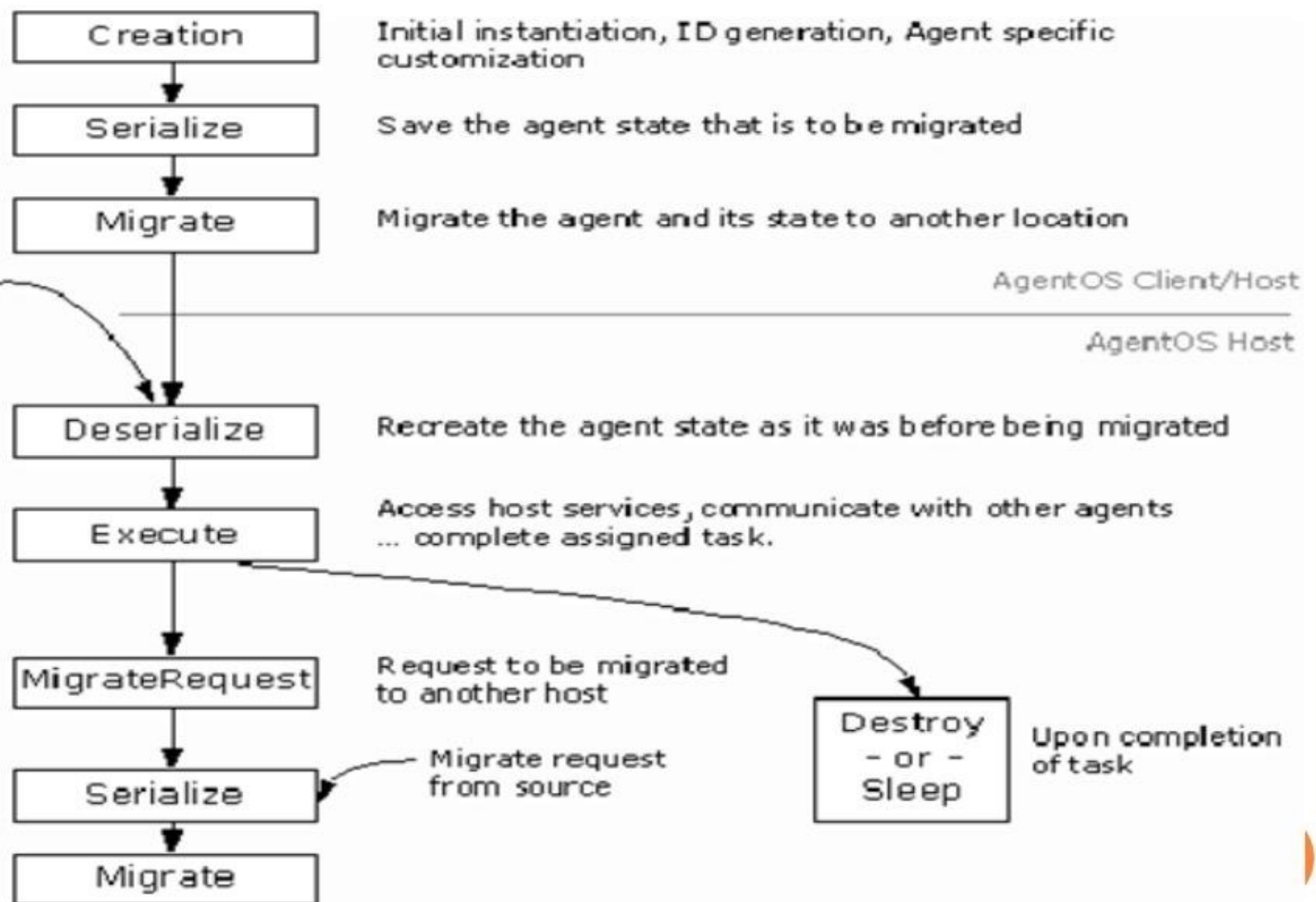


Figure 1 : The Agent Lifecycle

AGENT OS REQUIREMENTS

- Platform independence
- Dynamic agent invocation
- Migration of agents and their state
- Inter-agent communication



AGENT OS REQUIREMENTS(CONTD..)

- Agent reuse
- Unique naming scheme
- Security
- Feature-rich



DESIGN OF AGENT OS



SYSTEM COMPONENTS

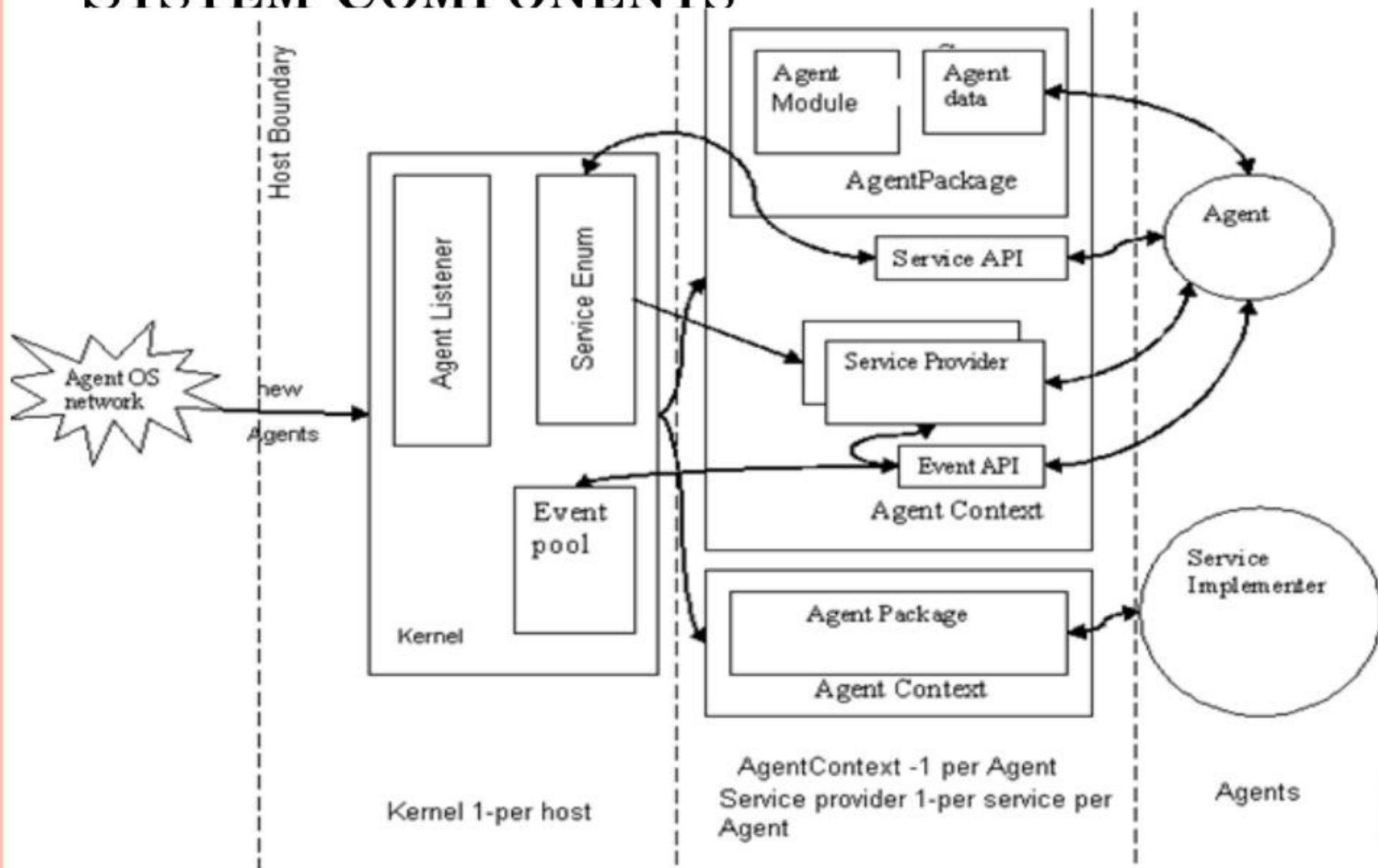


Figure 2 : Structural Overview of the AgentOS Architecture

SYSTEM COMPONENTS(CONTD..)

○ Kernel

- Provides access to system services
- Monitoring n/w connection
- Implement communication protocol

○ Service Enum

- Part of kernel subsystem
- Manages a list of standard and extended services



SYSTEM COMPONENTS(CONTD..)

○ Event Pool

- Each AgentOS s/m contains a single event pool
- Provides queue based communication b/w agents

○ Agent Context

- Provides execution model for agents
- Starts, stops and controls the execution of agents



SYSTEM COMPONENTS(CONTD..)

- Agents

- Implements computational logic

- AgentPackage

- Used in migration of the agent
- Contains agents & resources needed to execute agents

- Service Providers and Service Implementers

- Service provider - API for agent to access services
- Service implementer – implements actual service



INSTANCE IDS

○ AgentOSID

- Static ID
- Combination of n/w address, time stamp and a sequential counter
- Guarantees uniqueness to static objects

○ AgentInstanceID

- Guarantees uniqueness to static & active objects
- Combination of n/w address and time of creation



ADVANTAGES

- Load balancing & Fault tolerance
- Disconnected operation
- Reduced Inter-node communication
- Event based communication
- Secure transaction



CONCLUSION

- AgentOS is an OS “of agents, for agents, and by agents”
- System functions are carried out by agents, hence an OS of agent
- Agent OS manages the agents,hence OS for agents
- For agent communication underlying protocol is partially generated by agents



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ANY QUESTIONS???



THANK YOU...

